Grid Interconnection Application Form

Transmission connection — storage

PROJECT ID: TX-STO-003
PROJECT NAME: Pennine Battery Storage
APPLICANT: Pennine Flexible Power Ltd
LEVEL: transmission
CONNECTION TYPE: storage
CAPACITY: 200 MW / 800 MWh BESS
STATUS: Under review
DATE SUBMITTED: 2026-04-29

SECTION 1: Energy capacity

REQUIREMENT:
Energy storage capacity (MWh) and rated duration. Source: Grid Code Storage Code user definitions.

SUBMITTED:
Energy capacity 800 MWh at beginning of life, rated duration 4 hours at 200 MW full discharge. Lithium-ion LFP chemistry.

SUPPORTING DOCS:
none

SECTION 2: Charge / discharge

REQUIREMENT:
Maximum charge rate (MW import) and discharge rate (MW export); whether equal or asymmetric. Source: CUSC App. C.

SUBMITTED:
Maximum discharge (export) 200 MW; maximum charge (import) 200 MW. Symmetric bidirectional inverter rating.

SUPPORTING DOCS:
single_line_diagram.pdf

SECTION 3: Connection capacity registration

REQUIREMENT:
Both import capacity and export capacity declared separately. Source: CUSC §2.2.4 / §2.3.

SUBMITTED:
Export (TEC) registered at 150 MW. Import capacity registered at 200 MW.

SUPPORTING DOCS:
single_line_diagram.pdf

SECTION 4: Land — control

REQUIREMENT:
Evidence of land rights of at least 20 years. Source: Gate 2 §4.1c.

SUBMITTED:
25-year lease executed over the 4-hectare site adjacent to the existing 275 kV Stalybridge substation.

SUPPORTING DOCS:
land_lease_agreement.pdf

SECTION 5: Ancillary-service capability

REQUIREMENT:

Frequency response (dynamic/static), reactive power range, black-start capability declaration. Source: CUSC App. F3 / F4.

SUBMITTED:

Dynamic Containment and Dynamic Regulation capable across the full 200 MW. Reactive range +/- 0.95 power factor at the connection point. Black-start capable.

SUPPORTING DOCS:

none

SECTION 6: Dynamic stability

REQUIREMENT:

Inverter control parameters, grid-forming vs grid-following declaration, fault-ride-through curves. Source: Grid Code CC.6.3 (FRT).

SUBMITTED:

Grid-forming inverters with virtual synchronous machine control. FRT compliant to the Grid Code CC.6.3.15 voltage-against-time profile, including the 140 ms zero-voltage ride-through requirement.

SUPPORTING DOCS:

none